

Gauge-Covariant Correlator Closure for the q-D-H Portal

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral;
see RECOVERY_PROVENANCE.md.

Executive result

The arbitrary conventional off-shell elementary Higgs self-energy is gauge dependent. The corrected target is a PT/BFM hard-soft retarded kernel with Ward/ST-consistent vertices and a gauge-singlet $H^\dagger H$ control. Fermion and scalar Ward residuals are $1.53\text{e-}15$ and $7.07\text{e-}14$. Bare-vertex 2PI is rejected; three-loop 3PI or equivalent Bethe-Salpeter closure is the correct HPC target.

Reported benchmark values

- metric_1: 0.0011585159
- metric_2: $1.53\text{e-}15$
- metric_3: $7.07\text{e-}14$
- metric_4: $5.24\text{e-}16$
- metric_5: $1.42\text{e-}12$
- metric_6: 0.688
- metric_7: 0.838
- metric_8: 0.000944